Fraud Detection Model Through Global Structure and Local Substructure Feature Fusion

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***Abstract*—Financial fraud detection plays an imperative role in preserving the safety and trustworthiness of digital economies. Graph modeling, by representing accounts, merchants, and transactions as graph-structured data, enables fraud detection to capture hidden entity connections and structural interactions, but its effectiveness is restricted by severe class imbalance and disjoint information perspectives. To address these challenges, we propose TAHIA, a novel fraud detection model through global structure and local substructure feature fusion. Specifically, the global structure feature is optimized via feature similarity and propagation, where the feature mapping is enhanced by a gated dynamic attribute pattern module. Meanwhile, local scenario semantics are captured based on meta-paths, and heterogeneous features are aggregated across diverse node types. Finally, an attention-based multi-view fusion mechanism adaptively combines these complementary perspectives to achieve a robust fraud representation. Extensive experiments on multiple benchmarks demonstrate that TAHIA consistently achieves SOTA performance, validating its exceptional capability in overcoming severe class imbalance, bridging disjoint perspectives, and preserving topological diversity.**

***Index Terms*—fraud detection, heterogeneous network, scenario semantics, graph neural network, attention-based fusion**

I. Introduction

Fraud detection is a critical cornerstone of modern financial security. With the rapid growth of online transactions, financial systems are increasingly exposed to sophisticated and coordinated fraud behaviors, such as account takeover, synthetic identity theft, and cross-border money laundering [1]. These fraudulent activities not only increase the credit risk faced by financial institutions, but also disrupt the stability of financial markets [2]. As fraud strategies continue to evolve, intelligent and adaptive detection models are increasingly important for capturing complex fraud patterns in dynamic financial environments. Therefore, developing effective fraud detection models has become essential for protecting the security and trustworthiness of modern financial ecosystems.

Fraud detection increasingly relies on graph modeling to uncover hidden anomalous relationships. In graph-based fraud detection, financial entities such as accounts, merchants, devices, and users are represented as nodes, when transfers, payments, reviews, and shared behavioral relations are represented as edges. Existing studies have improved graph-based fraud detection from different perspectives. Camouflage-aware methods filter noisy or suspicious neighbors to reduce the influence of fraudsters deliberately connected with benign nodes [3]. Heterogeneous graph and meta-path methods model different entity types and relation paths to capture scenario-specific semantic dependencies [4]. Recent graph anomaly detection methods further enhance representation learning through self-supervised objectives, multi-space propagation for preserving anomaly-related distances [5], or attribute-pattern modeling for capturing non-linear abnormal feature intervals [6]. This formulation enables the model to capture entity connections and structural interactions beyond isolated transaction records. In a word, by shifting fraud detection from independent record classification to relational pattern mining, graph-based methods provide a more expressive way to identify concealed fraud behaviors. However, most of these methods still focus on specific signals and pay limited attention to the fusion of topologically diverse information. Consequently, class imbalance and disjoint perspectives remain persistent bottlenecks in graph-based fraud detection.

Fraud detection under class imbalance and disjoint perspectives motivates us to design a more comprehensive framework. To address these challenges, we propose TAHIA, a novel model through global structure and local substructure feature fusion, which integrates graph structural information, scenario-specific semantic information, and heterogeneous contextual information, while capturing topological diversity through dynamic attribute patterns. Specifically, TAHIA first constructs feature similarity graphs and feature propagation graphs to optimize structural representation, and introduces a GAAP-inspired dynamic attribute pattern mapping module to capture non-linear attribute interval patterns [6]. Then, scenario semantic features are extracted through meta-path-based representation learning with MP2Vec [4]. In addition, restart random walk and type-specific BiLSTM encoders are used to aggregate heterogeneous neighborhood information. Finally, an attention-based fusion mechanism combines multiple feature views to obtain the final fraud representation.

The main contributions of this paper are as follows:

- **Multi-view fraud detection framework:** We introduce a comprehensive fusion model that integrates structural, semantic, and heterogeneous perspectives of transaction data.

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- **Local substructural feature learning:** We integrate scenario-specific meta-path semantics, type-aware heterogeneous neighbors to enrich local fraud representations.
- **Robust Experimental Validation:** We demonstrate the superiority of our model across multiple benchmark datasets, validating its effectiveness and generalization capability.

II. Related Work

Financial fraud detection is commonly formulated as an anomaly detection problem. Early studies mainly relied on rule-based systems or traditional machine learning models to identify fraudulent behaviors from handcrafted features [7], [8]. Although these methods are effective for static and independent features, they are limited in modeling the complex topological dependencies in transaction networks. Therefore, recent studies have increasingly adopted graph-based methods for fraud detection.

Fraud-aware GNNs. Graph Neural Networks (GNNs) have been widely used to capture structural dependencies in financial networks [9]. Since fraudsters may deliberately connect with legitimate users to disguise themselves, several fraud-aware GNNs are designed to reduce noisy aggregation. CARE-GNN [3] filters camouflaged neighbors through a label-aware mechanism, PC-GNN [10] selects informative neighbors to alleviate class imbalance, and FRAUDRE [11] introduces fraud-aware graph convolution to handle inconsistency in fraud graphs. These methods improve fraud-specific neighborhood aggregation, but they pay limited attention to heterogeneous entity types and scenario-level semantics.

Heterogeneous Graph and Meta-Path Semantic Learning. Real financial networks usually contain multiple entity types and relation types, which motivates heterogeneous graph learning [12]. Early heterogeneous information network studies explored graph-based classification and semantic similarity, such as GNetMine [13], and HeteSim [14]. SemiGNN [15] and MAHINDER [16] model heterogeneous financial networks with attention-based mechanisms, while HACUD [17] exploits meta-path-based neighbors for cash-out user detection. In addition, metapath2vec [4], HINE [18], and TransPath [19] learn semantic representations for heterogeneous information networks through meta-path-guided or relation-aware embedding strategies. These methods capture node-type correlations and semantic path dependencies, but they usually do not jointly consider structural optimization, heterogeneous context, and attribute-level anomaly patterns.

Advanced Graph Anomaly Detection Paradigms. Recent graph anomaly detection methods further improve anomaly representation from broader perspectives. UniGAD [20] unifies node-, edge-, and graph-level anomaly detection through shared subgraph-level representations. SpaceGNN [5] projects nodes into multiple geometric spaces and performs distance-aware propagation to preserve anomaly-related signals. GAAP [6] models global attribute-association patterns by learning dynamic attribute intervals for graph fraud detection. More recent studies further extend graph fraud detection to practical settings, such as structure-aware heterogeneous relation modeling [21], temporal feedback learning on dynamic heterogeneous graphs [22], node pattern mining in social context [23], [24] and open-set unknown fraud detection with adaptive prototype learning [25]. Although these methods improve anomaly detection from multi-level, geometric, or attribute-based perspectives, they still tend to emphasize one dominant information source.

Despite these tremendous advancements, existing methods usually focus on a single perspective, such as neighborhood filtering, relation modeling, semantic paths, or attribute patterns, while the joint challenge of class imbalance and fragmented graph information remains insufficiently explored. Financial fraud in production scenarios, such as the monthly transaction dataset (e.g., CCCO), is inherently multi-scale, simultaneously reflected in structural anomalies, contextual behavior shifts, and heterogeneous interaction patterns. This limitation highlights the urgent need for a unified framework capable of adaptively fusing structural topology, scenario semantics, and heterogeneous interactions.

III. Task Definition

The objective of this task is to design a fraud detection model to detect fraudulent users from a heterogeneous network graph of financial transactions. This work represents the fraud detection task as a classification problem for account nodes [26]. The heterogeneous financial transaction network is represented as $G = \{V, E, X, A, Y, R, T\}$. The variables are defined as follows:

- $V = \{v_1, v_2, ..., v_N\}$ represents the set of nodes in the graph, where v_i denotes an individual node and N is the total number of nodes in the network. T represents the type of node. A mapping function $\tau : V \rightarrow T$ is employed to obtain node types, namely $\tau(v_i) \in T$. A node set of a certain type t is represented as V^t, while the node set to be detected is denoted as V^0 since their type id is 0.
- $X = \{X_1, X_2, X_3, ..., X_{|T|}\}$ represents the set of feature matrices for different node types, where $X_t \in \mathbb{R}^{|V^t| \times d_t}$. x_i denotes the static feature of node v_i and d_t is the static feature dimension for nodes of type t.
- $E = \{e_1, e_2, ..., e_M\}$ represents the set of edges, where e_i denotes an edge and consists of a head node and a tail node, e.g. $e_i = (v_k, v_t)$. The edge relationships formed by different types of head and tail nodes are various, so a mapping function $\phi : E \rightarrow R$, $\phi(e_i) \in R$ is employed to obtain the types of the edges. Edges under the same relationship r form an adjacency matrix, denoted as A_r, and $A = \{A_1, A_2, ..., A_{|R|}\}$.
- $Y = \{y_1, y_2, ..., y_{|V^0|}\}$ represents the labels of nodes, where $y_i \in \{0, 1\}$. A label of 0 indicates a normal node, while a label of 1 indicates a fraudulent node.

The input of the model is a heterogeneous network representation of financial transactions, denoted as $G = \{V, E, X, A, Y, R, T\}$. The output is the predicted label whether the node is a fraudulent node.

IV. METHODOLOGY

The proposed model consists of three features from the original transaction graph as shown in Fig. 1.

The graph structure feature is optimized by the generation of feature similarity and feature propagation, where the original feature mapping is enhanced by a gated dynamic attribute pattern module. The semantic feature is obtained by the scenario structure, which is defined based on meta-paths. The heterogeneous information feature is captured by aggregating features of different types of nodes. A multi-view feature fusion is finally employed to have the total features of graph. Based on the total features, the probability of fraud nodes are calculated based on the attention mechanism and classify layer.

A. Graph Structural Features

The structural view aims to optimize the original heterogeneous transaction graph by integrating feature similarity and node-type feature propagation. Given the raw feature $\mathbf{x}_i^{raw} \in \mathbb{R}^F$ of node v_i, a type-specific mapping is first applied:

$$\mathbf{h}_i^{lin} = W_{\tau(v_i)}\mathbf{x}_i^{raw} + \mathbf{b}_{\tau(v_i)}. \tag{1}$$

where $\mathbf{x}_i^{raw} \in \mathbb{R}^F$ denotes the raw feature vector of node v_i, F is the unified feature dimension after preprocessing, $W_{\tau(v_i)} \in \mathbb{R}^{D\times F}$ is a trainable type-specific transformation matrix, and $\mathbf{b}_{\tau(v_i)} \in \mathbb{R}^D$ is the corresponding bias term.

Inspired by GAAP [6], a gated dynamic attribute pattern encoder is introduced to enhance the feature mapping. Each normalized attribute value $\tilde{x}_{ij}$ is softly assigned to K learnable bins:

$$p_{ijk} = \frac{\exp\left(-\frac{(\tilde{x}_{ij}-c_k)^2}{s_k\tau}\right)}{\sum_{k'=1}^{K}\exp\left(-\frac{(\tilde{x}_{ij}-c_{k'})^2}{s_{k'}\tau}\right)}, \tag{2}$$

where c_k and s_k denote the learnable bin center and width, respectively. The dynamic attribute pattern representation is computed as:

$$\mathbf{h}_i^{dap} = \frac{1}{\sqrt{F}}\sum_{j=1}^{F}\sum_{k=1}^{K} p_{ijk}\mathbf{e}_{jk}. \tag{3}$$

where $\mathbf{e}_{jk} \in \mathbb{R}^D$ denotes the learnable embedding corresponding to the k-th bin of the j-th attribute. The normalization term $\sqrt{F}$ is used to stabilize the scale of the aggregated pattern representation. The final mapped representation is obtained through gated residual fusion:

$$\mathbf{x}_i = \sigma\left(\mathbf{h}_i^{lin} + \sigma_g(g)\mathbf{h}_i^{dap}\right), \tag{4}$$

where $\sigma(\cdot)$ is the ReLU activation function, $\sigma_g(\cdot)$ denotes the sigmoid function, and g is a learnable scalar gate. This gated residual design preserves the stability of the original feature mapping while introducing non-linear attribute interval patterns that are useful for fraud detection.

For each relation r, the feature similarity graph is generated by thresholding a weighted multi-head cosine similarity:

$$S_{r,ij}^{fs} = \begin{cases} \psi_r^{fs}(\mathbf{x}_i,\mathbf{x}_j), & \psi_r^{fs}(\mathbf{x}_i,\mathbf{x}_j) \geq \omega^{fs}, \\ 0, & \text{otherwise}, \end{cases} \tag{5}$$

Algorithm 1 Structural Graph Generation

Input: Raw feature matrix X^{raw}, relation adjacency matrices $\{A_r\}$, node type information $\tau(\cdot)$.
Output: Structural graphs $\{S_r^{stru}\}$.

```
for each node type t do
    Apply type-specific gated DAP mapping to obtain mapped features X_t.
end for
Concatenate all type-specific mapped features as X.
for each relation r do
    Extract the original relation graph A_r.
    Generate feature similarity graph S_r^fs using mapped features X.
    Compute head-side raw feature similarity S_r^fh and obtain S_r^fph = S_r^fh A_r.
    Compute tail-side raw feature similarity S_r^ft and obtain S_r^fpt = A_r (S_r^ft)^⊤.
    Fuse graphs by S_r^stru = Att([S_r^fs, S_r^fph, S_r^fpt]).
end for
```

$$\psi_r^{fs}(\mathbf{x}_i,\mathbf{x}_j) = \frac{1}{H}\sum_{h=1}^{H}\cos(\mathbf{w}_{h,r}^{fs}\odot\mathbf{x}_i, \mathbf{w}_{h,r}^{fs}\odot\mathbf{x}_j). \tag{6}$$

To further model indirect transaction relations, node-type feature propagation is performed on the head and tail sides of relation r:

$$S_r^{fph} = S_r^{fh}A_r, \qquad S_r^{fpt} = A_r(S_r^{ft})^\top, \tag{7}$$

where S_r^{fh} and S_r^{ft} are obtained by applying the same similarity function to raw features of head-type and tail-type nodes, respectively. The three structural graphs are then fused by channel-level attention:

$$S_r^{stru} = Att([S_r^{fs}, S_r^{fph}, S_r^{fpt}]). \tag{8}$$

where $Att(\cdot)$ denotes a channel-level graph attention operation. It first normalizes each candidate graph and then learns a softmax weight for each graph channel to obtain their weighted sum. Different from query-key-value attention, this operation focuses on adaptively balancing multiple generated graph structures.

The overall procedure of structural graph generation is summarized in Algorithm 1.

After obtaining S_r^{stru}, the structural graph is further combined with the scenario semantic graph and the original heterogeneous graph to obtain the optimized graph $\hat{S}$. A GCN encoder is then applied to learn structural representations:

$$X^{stru,(l+1)} = \sigma(\bar{S}X^{stru,(l)}W^l), \tag{9}$$

where $\bar{S}$ is the normalized form of $\hat{S}$. The output of the last GCN layer is denoted as X^{stru}.

B. Semantic Features

Different fraud scenarios usually correspond to different high-order interaction patterns in heterogeneous graphs. To capture scenario-specific semantics, we use meta-paths to

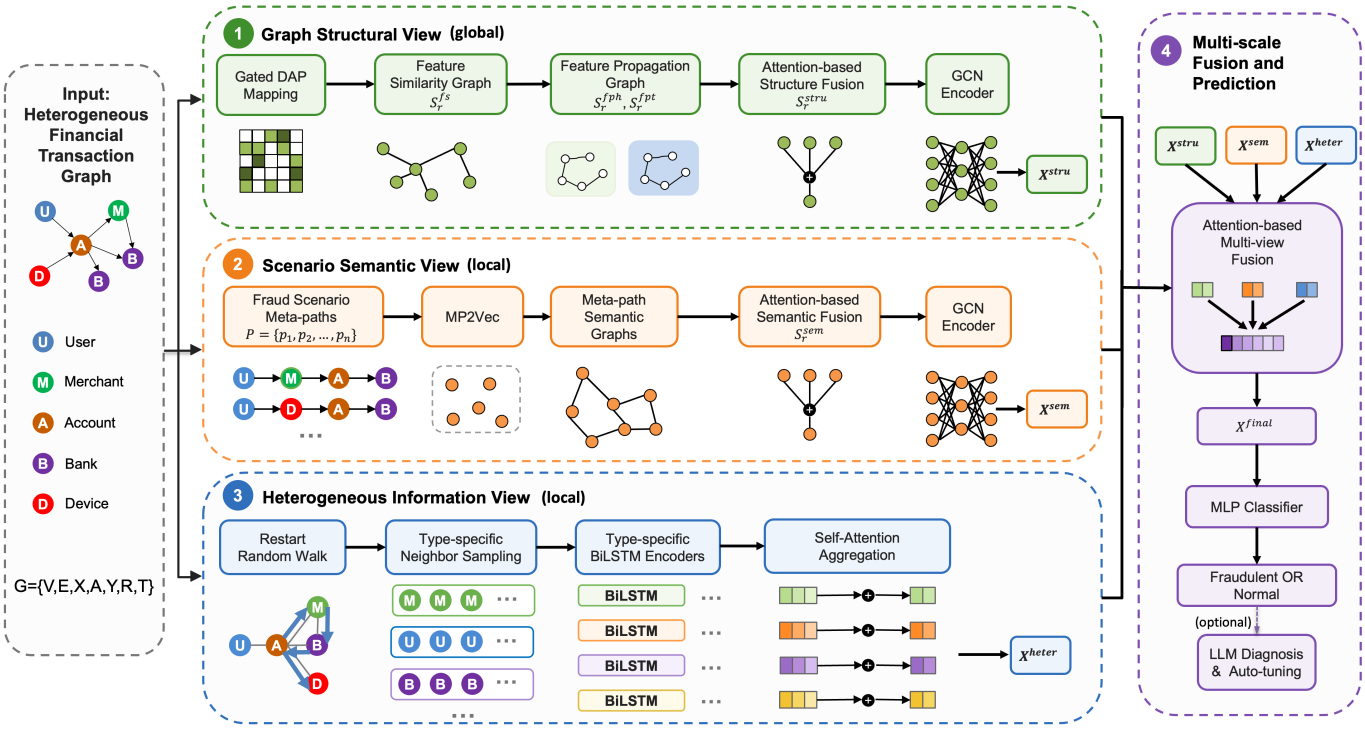


Fig. 1. Architecture of the Fraud Detection Model Integrating Heterogeneous Information and Scenario semantics

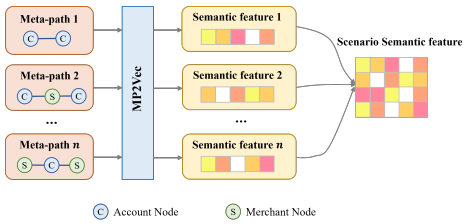


Fig. 2. Process of Generating Scenario Semantic Features

describe multi-hop relation sequences among different types of nodes. For example, the meta-path 'account-merchant-account' indicates that two accounts are connected through the same merchant, which may reveal indirect cash-out behavior. The process of generating scenario semantic features is shown in Fig. 2.

Meta-paths are defined based on fraud scenarios, denoted as $P = \{p_1, p_2, ..., p_n\}$. Directly integrating adjacency matrices along meta-paths may cause high computational cost and information loss [27]. Therefore, MP2Vec [4] is adopted to learn meta-path embeddings, denoted as $F_p = \{F_{p1}, F_{p2}, ..., F_{pn}\}$. Then, a semantic adjacency matrix $S^p_{r,ij}$ for each meta-path is obtained by Eq. (10):

$$S^p_{rk,ij} = \begin{cases} \psi^p_{r,k}(f^k_i, f^k_j) & \psi^p_{r,k}(f^k_i, f^k_j) \geq \omega^p \\ 0 & other \end{cases} \tag{10}$$

where f^k_i and f^k_j are the i-th and j-th rows of F_{pk}, and $\psi^p_{r,k}$ follows the same calculation as Eq. (6). The semantic adjacency matrices of different meta-paths are fused into the scenario semantic graph under relationship r:

$$S^{sem}_r = Att([S^p_{r1}, S^p_{r2}, ..., S^p_{rn}]), \tag{11}$$

where $Att(\cdot)$ calculates the importance weights of different meta-path-based semantic graphs. Finally, the scenario semantic graph is fed into a GCN encoder.

Finally, the scenario semantic graph is fed into a GCN encoder to obtain the semantic feature representation:

$$X^{sem,(l+1)} = \sigma\left(\bar{S}^{sem} X^{sem,(l)} W^l_{sem}\right), \tag{12}$$

where $\bar{S}^{sem}$ denotes the normalized form of the fused scenario semantic graph, and W^l_{sem} is a trainable parameter. The output of the last semantic GCN layer is denoted as X^{sem} and used for subsequent multi-view fusion and classification.

C. Heterogeneous Information Features

In the neighborhood of a certain node, the distribution of different types of nodes is uneven. In order to obtain information on various types of nodes, the restart random walk [28] is employed to select nodes from the node neighborhood, as shown in Figure 3.

Given a walk length L, a node list is obtained through L steps of random walk or restart based on the adjacency matrix, as shown in Eq. (13):

$$Neibor(v_i) = \{..., |\{v_j \in V^t\}|_{=k_t}, ...\} \tag{13}$$

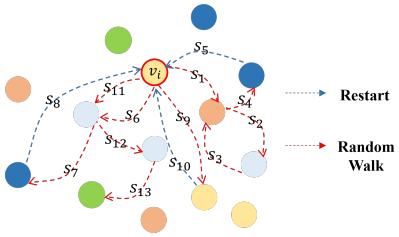


Fig. 3. Process of Node Selection via Restart Random Walk

where k_t represents the number of nodes of each type. After that, a BiLSTM encoder is used to encode each type of node separately. Different encoders emphasize distinct aspects, allowing for the enhancement of information diversity and a more accurate representation of nodes, as shown in Eq. (14):

$$H_t = BiLSTM_t(\{\mathbf{x}_j \mid v_j \in Neibor_t(v_i)\}) \tag{14}$$

where $\mathbf{x}_j$ denotes the mapped feature representation of node v_j. The type-specific neighbor representation is obtained by mean pooling over all time-step outputs:

$$Z_t = \text{MeanPool}(H_t). \tag{15}$$

All type-specific representations are denoted as $Z = \{Z_0, Z_1, ..., Z_{|T|-1}\}$, where Z_0 denotes the mapped representation of the target node and Z_t denotes the BiLSTM-based representation of the t-th type-specific neighbor sequence.The BiLSTM consists of forward and backward outputs, with the same process. The calculation process of the forget gate is as follows:

$$f_l = \sigma(W_{fx}x_l + W_{fh}h_{l-1} + b_f) \tag{16}$$

where W_fx, W_fh, and f_q are trainable parameters. The input gate i_l and output gate o_l are calculated as Eq. (16) with different parameters. The hidden state h_l of this time step is calculated in Equations Eq. (17)-Eq. (19):

$$\tilde{c_l} = \tanh(W_cx\tilde{x_l} + W_{ch}h_{l-1} + b_q) \tag{17}$$

$$c_l = f_l \odot c_{l-1} + i_l \odot \tilde{c_l} \tag{18}$$

$$h_l = o_t \odot \tanh(c_l) \tag{19}$$

where $\tilde{c_l}$ is the candidate memory. The forward and backward outputs can be represented as h_l^f and h_l^b , respectively. The BiLSTM output is formed by concatenating forward and backward outputs, which is represented as $H_t = [h_{l_2}^f, h_{l_2}^b]$,where L_2 is the number of BiLSTM layers.

A channel-level attention mechanism is then used to aggregate the representations of different node types. The attention weight of the t-th type-specific representation is calculated as:

$$\alpha_t = \frac{\exp(a_t)}{\sum_{t'=0}^{|T|-1}\exp(a_{t'})}, \tag{20}$$

where a_t is a learnable attention parameter for the t-th type-specific representation.The aggregated feature representation is denoted as X^{Heter}.

$$X^{Heter} = \sum_{t=0}^{|T|-1} \alpha_t Z_t. \tag{21}$$

Finally, the heterogeneous information feature X^{Heter} is obtained.

D. Multi-Scale Feature Fusion

In the previous sections, three feature representations X^{stru}, X^{sem} and X^{heter} are obtained, which represent the structural features of the nodes, the semantic features, and the heterogeneous information features respectively. Each of these features captures different aspects of the node characteristics at varying scales. However, there may be implicit relationships between these multi-scale features. The structural and semantic information is first integrated into the graph-enhanced representation through graph-level attention, and then fused with the heterogeneous information representation for final prediction. Therefore, a graph-level attention mechanism is employed to calculate the importance of each multi-scale feature, following a process similar to Eq. (20)-Eq. (21). Based on the importance scores, the final node feature representation X^{final} is obtained. Subsequently, a feedforward neural network (MLP) is employed as the classifier to determine whether a node is fraudulent based on X^{final}.

$$\hat{\mathbf{y}} = \text{softmax}\left(MLP(X^{final})\right), \tag{22}$$

where $MLP(\cdot)$ denotes the feedforward classifier, and $\hat{\mathbf{y}} \in \mathbb{R}^2$ represents the predicted probability distribution over normal and fraudulent classes. The fraud probability is obtained from the component corresponding to the fraudulent class.

E. Optimization Strategy

In fraud detection, the imbalance between fraudulent and normal nodes may bias the model toward the majority class. To alleviate this problem, we adopt a class-weighted optimization strategy during training. Let M_0 and M_1 denote the numbers of normal and fraudulent nodes in the training set, respectively. The class weight coefficient is defined as $\alpha = M_0/(M_0+M_1)$, and the weights of the normal and fraudulent classes are set as $w_0 = 1-\alpha$ and $w_1 = \alpha$, respectively. Since fraudulent nodes are usually the minority class, this setting assigns a larger weight to fraudulent samples.

The weighted cross-entropy loss is then defined as:

$$\mathcal{L} = -\frac{1}{N}\sum_{i=1}^{N}\left[w_0\mathbb{I}(y_i=0)\log\hat{y}_{i,0} + w_1\mathbb{I}(y_i=1)\log\hat{y}_{i,1}\right], \tag{23}$$

where $\hat{y}_{i,0}$ and $\hat{y}_{i,1}$ denote the predicted probabilities of node v_i belonging to the normal and fraudulent classes, respectively, and $\mathbb{I}(\cdot)$ is the indicator function.

V. Experiments

In this section, we conduct extensive experiments to evaluate the effectiveness of TAHIA. The experiments are organized into three parts. First, we present the experimental setup,

including the datasets, baselines, parameter settings, and evaluation metrics. Second, we report the performance comparison results, where TAHIA is compared with a series of state-of-the-art baselines to demonstrate its effectiveness in graph-based fraud detection. Finally, we provide an analysis of TAHIA through ablation studies and parameter sensitivity analysis to investigate the contribution of different modules and validate the overall design choices.

A. Experimental Setup

1) Datasets: We evaluate TAHIA on four datasets, including one private credit card cash-out dataset, CCCO[1], and three public datasets, DBLP, ACM, and AMAZON. CCCO is constructed from simulated credit card cash-out transactions. It contains two types of nodes, account nodes and merchant nodes. Edges between account nodes indicate transfers, while edges between account and merchant nodes indicate consumption. Fraudulent account nodes are labeled according to abnormal cash-out behaviors involving frequent card swipes, merchant collusion, and cyclic transaction patterns. DBLP and ACM are citation network datasets with multiple node types and relation types, which are adopted from the heterogeneous graph benchmarks used in [29], respectively. Following graph anomaly detection settings, their original labels are converted into binary anomaly labels. Amazon dataset lays emphasis on identifying anomalous users who are paid to write fake reviews in the Musical Instrument category on Amazon.com, being adopted from the graph-based fraud detection benchmark used in [3]. For all datasets, only labeled target nodes are used for supervised training and testing, while the remaining graph structure is retained to provide contextual information when applicable. The labeled target nodes are split into training and testing sets with a ratio of 10% to 90%. The statistics, node types, meta-paths, and sample distributions are shown in Tables I–III. In CCCO, C and S denote account and merchant nodes, respectively. In DBLP, A, P, and C denote author, paper, and conference nodes. In ACM, P, A, and S denote paper, author, and subject nodes. In AMAZON, U denotes user nodes.

2) Baseline Models: We compare TAHIA with six representative baseline models. GCN [30] is a classical homogeneous graph neural network model. CARE-GNN [3] enhances fraud detection by filtering camouflage neighbors with a label-aware similarity mechanism. FRAUDRE [11] models fraud-aware graph relations to alleviate inconsistency and class imbalance in fraud detection. UniGAD [20] unifies multi-level graph anomaly detection by jointly considering anomaly signals from different graph levels. SpaceGNN [5] captures node anomaly patterns from multiple latent spaces under extremely limited labels. GAAP [6] models dynamic attribute patterns to improve graph anomaly detection.

3) Parameter Settings: The main parameter settings of TAHIA are reported in Table IV. For all datasets, the model is trained for 300 epochs with a learning rate of 0.005. The sparsity thresholds for feature similarity, feature propagation, and semantic graph generation are adjusted according to the graph density and feature characteristics of each dataset. Other shared settings, including the number of GCN layers, BiLSTM layers, and attention heads, are kept consistent across datasets when applicable. All experiments use a single NVIDIA 4090 GPU with 24GB memory.

4) Evaluation Metrics: The dataset statistics indicate that all four datasets exhibit category imbalances. To ensure a fair evaluation of the fraud detection performance, the evaluation metrics must not favor any particular category. Therefore, the overall performance of the model is assessed using three key metrics: precision (P), recall (R), and F1 score (F1-Score, $F1$). Additionally, unweighted averages are calculated for these metrics across different categories to derive macro precision (Precision-Macro, $Pre.$), macro recall (Recall-Macro, $Rec.$), and macro F1-Score (F1-Macro, $F1.$), which serve as the final evaluation metrics.

TABLE I
DATASET STATISTICS

Dataset	Nodes(Feat.)	Types	Edges	Anom.(%)
CCCO	13351(32)	2	504832	5.81
DBLP	7305(334)	3	19816	7.51
ACM	8994(1902)	3	25922	10.78
AMAZON	11944(25)	1	8847096	6.87

TABLE II
NODE TYPES AND META-PATHS FOR EACH DATASET

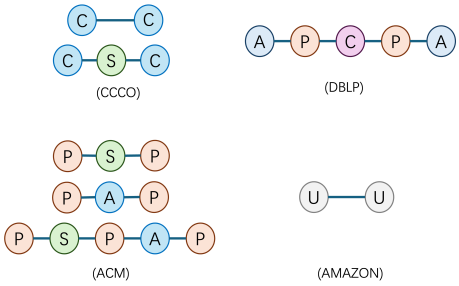


TABLE III
SAMPLE DISTRIBUTION OF LABELED TARGET NODES IN TRAIN AND TEST SETS

Dataset		Fraud Nodes	Normal Nodes
CCCO	Train Set	77	1228
	Test Set	690	11061
DBLP	Train Set	55	240
	Test Set	494	2168
ACM	Train Set	97	205
	Test Set	873	1850
AMAZON	Train Set	82	1112
	Test Set	739	10011

B. Performance Comparison

This section provides a comprehensive evaluation of the proposed TAHIA model. Our model is compared with baseline

[1]The data and proposed method are available at https://github.com/fieldchaser/TAHIA-Fraud-Detection-Model/tree/main/mycode/TAHIA

TABLE IV
PARAMETER SETTINGS FOR TAHIA

Param.	CCCO	DBLP	ACM	AMAZON
Epochs	300	300	300	300
Learning Rate	0.005	0.005	0.005	0.005
Dropout	0.2	0.2	0	0.2
ω_{fs}	0.1	0.99	0.8	0.1
ω_{fp}	0.1	0.99	0.2	0.1
ω_{sem}	0.1	0.4	0.6	0.1
GCN layers(L_1)	2	2	2	2
BiLSTM layers(L_2)	1	1	1	1
Hidden Space Dimensions	64	256	32	64
Heads	2	2	2	2

models on one private dataset (CCCO) and three public datasets (DBLP, ACM and AMAZON). The results are shown in Fig. 4.

As shown in Fig. 4, TAHIA achieves the best overall performance on the four datasets, especially in terms of $F1$., which is the main evaluation metric in this work. On ACM, TAHIA outperforms the second-best model by 0.54%, 2.05%, and 1.36% in $Pre.$, $Rec.$, and $F1.$, respectively. On DBLP, the improvements are more significant, with gains of 4.40%, 3.26%, and 3.92% on the three metrics. On CCCO, although the precision is 1.27% lower than GAAP, TAHIA achieves higher $Rec.$ and $F1.$, with improvements of 0.36% and 1.45%, respectively. On AMAZON, TAHIA improves $Pre.$ and $F1.$ by 3.83% and 1.18%, while its recall is slightly lower than FRAUDRE by 1.18%. These results indicate that TAHIA obtains a better balance between precision and recall and achieves more stable $F1.$ performance across different graph scenarios.

From the perspective of method categories, the general GNN model GCN performs poorly on CCCO and AMAZON, because it mainly relies on neighborhood aggregation and lacks fraud-specific mechanisms for class imbalance and abnormal behavior modeling. On DBLP and ACM, GCN obtains relatively competitive results, which is because citation networks usually have clearer community structures and smoother label distributions than fraud transaction datasets. However, its performance is still lower than TAHIA, showing that simple graph convolution is insufficient while facing some challenges such as class imbalance and separated information views.

Fraud-oriented models, including CARE-GNN and FRAUDRE, generally perform better than GCN on fraud-related datasets. CARE-GNN achieves strong recall on CCCO and competitive performance on AMAZON, indicating that neighbor filtering is useful for handling camouflaged fraudsters. FRAUDRE also improves recall on AMAZON by modeling fraud-aware relations. However, these models are less effective on DBLP and ACM, where anomaly patterns are more related to heterogeneous semantic structures rather than only fraud-specific neighbor selection or relation inconsistency. This suggests that methods designed for a single fraud pattern may not generalize well across different graph scenarios.

Advanced graph anomaly detection models, including UniGAD, SpaceGNN and GAAP show stronger generalization than traditional fraud-specific models on some datasets. The relatively lower performance of UniGAD may be due to differences in dataset preprocessing and evaluation protocols from its original setting. SpaceGNN performs well on ACM and AMAZON, which indicates that multi-space representation learning is helpful when anomaly signals are distributed in different latent spaces. Nevertheless, its performance on CCCO is limited, suggesting that distance-aware representation alone may not fully capture transaction-specific fraud semantics. GAAP achieves strong results on CCCO and DBLP, especially in $Pre.$ and $F1.$, because dynamic attribute pattern modeling is effective for detecting non-linear abnormal attribute intervals. However, GAAP mainly focuses on attribute patterns and does not explicitly integrate scenario semantic paths and heterogeneous neighborhood behaviors.

Compared with these baseline models, TAHIA combines graph structural optimization, scenario semantic modeling, heterogeneous information aggregation, and gated dynamic attribute pattern mapping. This design allows the model to capture both association patterns and attribute patterns, which is important for fraud detection under sparse labels and imbalanced classes. By simultaneously taking multiple distinct features into consideration, our framework effectively prevents the representation dilution commonly suffered by single-perspective models. The advantage is particularly clear on CCCO, where fraud behaviors are closely related to both transaction structures and abnormal attribute intervals. On DBLP and ACM, the improvement mainly comes from the joint use of structural and semantic information, while on AMAZON, the gain over GAAP shows that TAHIA can still benefit homogeneous fraud graphs through structural optimization and attribute pattern enhancement. The F1. comparison figures further confirm this trend by showing that TAHIA consistently outperforms representative baselines, including GCN, CARE-GNN, SpaceGNN, and GAAP, on all datasets.

C. Analysis of TAHIA

This subsection provides an in-depth analysis of the proposed TAHIA model. First, we conduct a progressive ablation study by gradually removing key components from the full model, including DAP mapping, semantic view, heterogeneous information view, and structural view. This setting helps evaluate not only the individual contribution of each module, but also the cumulative effect of combining multiple views. Then, we perform a sensitivity analysis on the initialization value of the DAP gate to investigate how the strength of dynamic attribute pattern injection affects model performance.

1) Ablation Study: To further evaluate the contribution of each component, we conduct a progressive ablation study by gradually removing key modules from the full TAHIA model. Since $F1.$ is the primary evaluation metric in this work and better reflects the balance between precision and recall under imbalanced labels, only $F1.$ scores are reported in this ablation study. Different from removing only one component independently, this progressive setting can better reflect how the model performance changes when the representation capacity

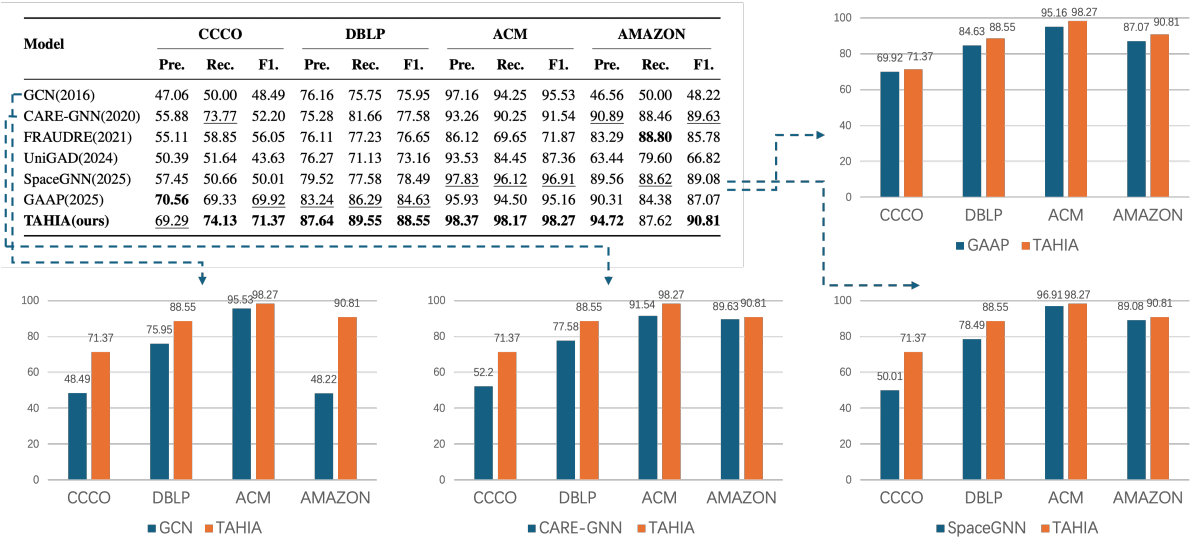


Fig. 4. Visible results of comparison experiment

is weakened step by step. Specifically, we first remove the DAP mapping, then remove the semantic view, heterogeneous information view, and structural view successively. This design helps examine both the effect of each module and the cumulative contribution of multi-view representation learning. The results are reported in Table V, where D, S, H, and T stand for DAP, semantic view, heterogeneous information view, and structural view, respectively.

TABLE V
ABLATION STUDY OF TAHIA

Model	CCCO	DBLP	ACM	AMAZON
Full	**71.37**	88.55	**98.27**	**90.81**
w/o D	68.12	**88.64**	98.13	89.31
w/o D+S	67.69	77.06	95.96	88.13
w/o D+S+H	69.69	79.27	95.89	90.32
w/o D+S+H+T	50.33	79.73	94.91	63.29

The ablation results show that the full TAHIA model achieves the best $F1.$ on CCCO, ACM, and AMAZON, and obtains the second-best result on DBLP with only a slight gap of 0.09% compared with w/o D. On CCCO, the performance drops from 71.37% to 68.12% after removing DAP, and further decreases to 67.69% after removing both DAP and the semantic view. When all major views are removed, the $F1.$ decreases sharply to 50.33%, showing that attribute pattern modeling and multi-view graph information are important for the private fraud scenario. This is consistent with the characteristics of credit card cash-out fraud, where abnormal behaviors are reflected in both transaction associations and non-linear attribute intervals. On ACM, the full model achieves the highest $F1.$, but the performance gap among several variants is relatively small. This is because the ACM graph has a clearer citation structure and is less complex than the CCCO transaction graph, so the original graph with GCN aggregation can already capture strong discriminative signals.

The results on DBLP and AMAZON show more dataset-dependent behavior. On DBLP, removing DAP slightly improves $F1.$, suggesting that the citation semantics and heterogeneous neighborhood information are already highly informative, while additional attribute pattern modeling may introduce limited redundancy. However, once the semantic view is removed, the $F1.$ drops significantly from 88.64% to 77.06%, confirming the importance of meta-path semantics for DBLP. On AMAZON, the full model achieves the best $F1.$, while w/o D+S+H remains close. Different from ACM, AMAZON has only one node type but contains much denser user-user connections and more complex fraud-related association patterns. Therefore, the original graph alone is easily affected by noisy neighborhood information. When all major enhancement modules are removed and only original graph aggregation is retained, the $F1.$ drops sharply to 63.29%. Overall, the progressive ablation results demonstrate that TAHIA benefits from the joint modeling of dynamic attribute patterns, scenario semantics, heterogeneous information, and graph structure, while the relative importance of each component varies across datasets.

2) DAP Sensitivity Analysis: To further investigate the influence of the gated dynamic attribute pattern encoder, a sensitivity analysis is conducted on the initial value of the DAP gate. The DAP gate controls the contribution of the dynamic attribute pattern representation in the feature mapping stage. Specifically, we vary the initial gate value from -5.5 to -3.5 while keeping all other model parameters unchanged. The results are shown in Table VI.

The sensitivity results show that the optimal DAP gate initialization varies across datasets. ACM achieves the best performance when the gate is initialized to -5.0, while DBLP, CCCO, and AMAZON obtain the best results at -4.5, -4.0, and -3.5, respectively. This indicates that the contribution

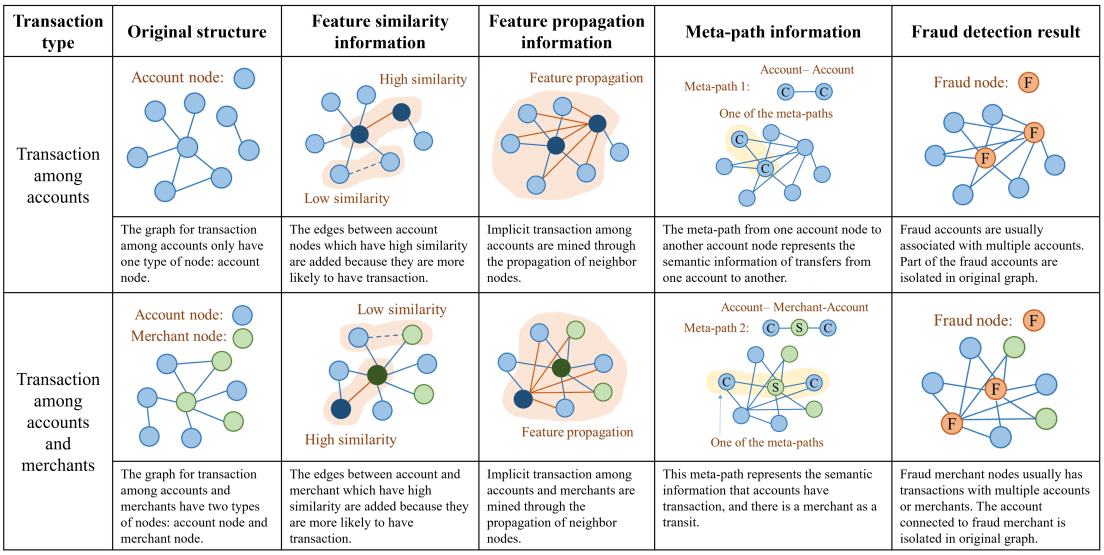


Fig. 5. Case study of different transaction scenario

TABLE VI
SENSITIVITY ANALYSIS OF DAP GATE INITIALIZATION

DAP Gate Init.	CCCO	DBLP	ACM	AMAZON
−5.5	66.65	87.68	98.01	90.12
−5.0	67.27	87.43	**98.27**	90.03
−4.5	63.83	**88.55**	98.19	90.20
−4.0	**71.37**	87.79	97.83	90.52
−3.5	68.68	87.39	97.53	**90.81**

of dynamic attribute patterns should be adjusted according to dataset characteristics. For citation-style datasets such as ACM and DBLP, a smaller gate is more suitable because their semantic and structural information is already highly informative, and excessive attribute pattern injection may introduce noise. In contrast, fraud-oriented datasets such as CCCO and AMAZON benefit from a relatively stronger DAP branch, since abnormal behaviors are often reflected in non-linear attribute interval patterns. Overall, the results confirm the necessity of using a gated residual design, which allows the model to balance the original feature mapping and dynamic attribute pattern enhancement.

3) Discussion on LLM-guided Diagnosis: We also explored a lightweight LLM-guided diagnosis module as an auxiliary training feedback tool. In our implementation, Qwen-Turbo is used as the LLM backend. After the first training stage, the module analyzes validation metrics, loss curves, class distribution, attention weights, and DAP gate values, and generates bounded suggestions for adjusting the DAP gate initialization or class-weighted loss. The LLM does not change the model architecture or directly participate in fraud classification. In our experiments, it provides interpretable diagnostic feedback, but its suggested updates do not consistently improve performance across all datasets. Therefore, we treat it as an auxiliary diagnosis component, and leave more reliable LLM-based fraud scenario explanation and training feedback for future work.

D. Case study

In this section, examples of how our approach proceed the original graph structure for two trading scenarios and the typical fraud nodes are presented. The meaning of each step for the transaction network and the characteristics of the fraud nodes are explained. The case is showed in Fig. 5.

For account-account transactions, the graph contains only account nodes. The feature similarity graph adds edges between accounts with similar transaction behaviors, such as transaction frequency and amount. The feature propagation graph further mines implicit account-account relations by propagating neighborhood information through similar nodes. Meanwhile, the meta-path C-C provides semantic information of direct transfers between accounts. By combining these signals, TAHIA can identify both fraud accounts associated with multiple accounts and fraud accounts that are relatively isolated in the original graph.

For account-merchant transactions, the graph contains both account nodes and merchant nodes. Feature similarity connects accounts and merchants with similar transaction characteristics, while feature propagation mines implicit account-merchant relations through neighboring nodes. The meta-path C-S-C is defined as a transaction path from account node to another account node with a merchant as the transit. This helps reveal fraud merchants with transactions involving multiple accounts or merchants, as well as isolated accounts connected to suspicious merchants.

VI. CONCLUSION

Class imbalance and disjoint information perspectives remain critical bottlenecks in graph-based fraud detection, especially when fraud behaviors are hidden within diverse transaction topologies. To overcome these limitations, this paper

introduces TAHIA, which simultaneously encodes structural topologies, scenario-specific semantics, heterogeneous contextual sequences, while adaptively embedding dynamic attribute patterns within financial transaction networks. By aligning these complementary signals, the proposed framework learns unified entity representations that better reflect complicated fraud behaviors. Extensive experiments confirm our method's SOTA performance, demonstrating that TAHIA outperforms existing methods in detecting fraudulent transactions. Notably, TAHIA achieves the best performance on CCCO, the private credit card cash-out dataset most closely aligned with the financial fraud scenario studied in this paper, further verifying its effectiveness in practical transaction-based fraud detection. However, the inherent latency and structural volatility of financial networks make long-term dynamic anomaly patterns particularly difficult to mine. In future work, we will further explore temporal fraud evolution modeling and more reliable LLM-assisted fraud scenario analysis.

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